Predicting Bank Customer Churn with Machine Learning: Hyperparameter Optimization and Class Imbalance Strategies

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Abstract

In the banking industry, retaining customers has become a critical strategic goal, largely due to escalating market competition and the significant expense associated with attracting new clients. This paper addresses the challenge of predicting customer churn through a systematic evaluation and optimization of various machine learning models. Using a publicly available Kaggle dataset, a preprocessing was applied, including class imbalance handling via SMOTE and ADASYN, and hyperparameter tuning through AutoML tools such as Optuna and TPOT. The findings reveal that ensemble models outperform traditional methods. Specifically, a LightGBM (LGBM) optimized by Optuna achieving the highest predictive power, reaching an Area Under the Curve (AUC) of 0.941 in the Kaggle competition. However, for business objectives that prioritize identifying the minority class, a Balanced Random Forest (BRF) classifier demonstrated superior performance on metrics relevant to imbalanced data, such as Recall (0.856) and G-Mean (0.86). The interpretability analysis using SHAP identified the number of products, customer age, and active membership status as the most influential predictors. Finally, we found that while Gradient Boosting models offer maximum predictive accuracy, the optimal model choice depends on whether the business goal is overall precision or effective identification of at-risk customers.

Keywords: Bank churn prediction, Machine learning, Ensemble models, Class imbalance, Hyperparameter optimization, SHAP

1. Introduction

1.1. Banking Customer Retention

Customer retention has become a fundamental strategic imperative for financial institutions, directly impacting their profitability and long-term revenue stability [1, 2]. In a banking market characterized by increased competition and greater fluidity in consumer behavior, loss or churn of customers represents a significant threat to organizational viability [3, 2].

Recent studies confirm the need to improve retention strategies in the banking sector. Compared to other sectors such as media (84%) or information technology (81%) the banking industry shows a weaker customer retention rate at 75% [2]. This disparity highlights a particular vulnerability of the banking sector, suggesting that the inherent nature of financial services may contribute to this lower retention rate.

The strategic approach has evolved significantly, shifting from prioritizing the acquisition of new customers to increasing investment in retaining existing ones. This reorientation is economically justified: acquiring a new customer is reportedly up to five times that of retaining a current one [4, 1]. According to [2], the proactive identification of at-risk customers enables banks to improve resource distribution and boost the overall effectiveness of their retention efforts.

1.2. A Machine Learning Approach to Churn Prediction

Recently, machine learning has been establishing itself as a robust tool for understanding and anticipating customer behavior [3, 5]. ML models excel in identifying complex patterns as well as the underlying factors that contribute to customer churn, even within large and heterogeneous datasets [3, 4].

Supervised machine learning allows the development of models that not only classify customers as churn or non-churn, but also assign a churn probability to each [3, 6]. This probabilistic capability is essential for customer service teams to implement targeted and personalized interventions [3, 6].

Compared to conventional approaches, machine learning optimization offers significant advantages in classifying churn risk, resulting in a predictive system with improved accuracy and increased efficiency [1]. ML models have the ability to determine not only whether a customer will churn, but also the probability of churn and the contributing factors, allows banks to design highly personalized and optimized retention strategies [2, 1].

1.3. Comparison of Classical and Advanced Models

In the literature on bank churn prediction, the use of machine learning models has increased, from classical statistical approaches to advanced ensemble and deep learning models. Due to their simplicity and transparent nature, classical models like Logistic Regression and Decision Trees are commonly employed as performance benchmarks.[3, 4].

However, advanced ensemble methodologies have consistently demonstrated superior performance. Random Forests are recurrently identified as a robust model, achieving high accuracy (97% [3], 99% [3], 85.9% [4], and 96% after tuning [7]). Their capacity for managing multicollinearity and high-dimensional data makes them particularly suitable for predicting customer churn [4].

Gradient Boosting algorithms, such as XGBoost, LGBM, and CatBoost, frequently emerge as the best-performing models or perform comparably to Random Forests [8, 1]. In [1], it was found that LGBM achieved the highest accuracy value of 0.8789, along with an AUC of 0.9694, outperforming other algorithms. XGBoost consistently demonstrates strong performance, with accuracy scores between 84% and 96% and AUC values from 0.9 to 0.93 across different research [8, 9, 7]. Table 1 presents a summary of the results obtained by some of these models in various studies, highlighting their effectiveness under different evaluation metrics. It is essential to interpret these results carefully, as they come from studies using different datasets, each with its own unique characteristics and distributions. Consequently, the table should be viewed as a representative overview of the state of the art, providing a useful benchmark rather than a direct comparison of algorithms.

Other algorithms, such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN), have also been examined. In fact, when comparing seven different algorithms, the study by [10] identified Gradient Boosting as the top-performing model, achieving an accuracy of 85.2% and an AUC of 0.87, highlighting its robustness for churn prediction.

The most commonly used evaluation metrics for assessing the performance of ML models are AUC-ROC, Accuracy, Recall, and F1-score. [1, 2, 7]. Hyperparameter tuning and techniques such as SMOTE are crucial for optimizing model performance, especially when dealing with imbalanced datasets [11, 7].

Table 1: Model performance in the literature for bank churn prediction.

Model	Precision (%)	Recall (%)	F1-score (%)	AUC	Reference	Dataset
Random Forest	87.0	98.0	92.0	0.698	[3]	Kaggle
Random Forest	85.9	—	—	0.852	[4]	Kaggle
Random Forest	87.0	88.0	88.0	—	[7]	Local
XGBoost	87.0	51.0	64.0	0.920	[8]	Local
XGBoost	83.0	84.0	84.0	—	[9]	Kaggle
XGBoost	90.0	85.0	87.0	—	[7]	Local
XGBoost	—	60.1	61.3	0.847	[12]	Kaggle
LGBM	89.8	85.5	87.6	0.969	[1]	Kaggle
Gradient Boosting	85.2	—	—	0.870	[2]	Kaggle
AdaBoost	90.0	66.0	76.0	0.930	[8]	Local
Logistic Regression	76.9	—	—	0.793	[4]	Kaggle

1.4. Sociodemographic and Transactional Tabular Data

The prediction of bank customer churn is based on tabular datasets comprising sociodemographic and transactional features [6, 7]. According to [4, 7], sociodemographic data includes attributes such as age, gender, country of residence, credit score, marital status, educational level, and estimated salary. Transactional data encompasses details such as active membership status, account balance, number of products used, credit card ownership, transaction count, and total transaction value [6, 7].

A widely used dataset is the Bank Customer Churn Prediction from Kaggle, which is synthetically generated from real-world data and serves as a common benchmark for binary classification tasks [6, 13]. The use of synthetic data promotes accessibility, reproducibility of results, and comparative evaluation across studies.

Several studies identify key features that significantly influence churn. The customer's age is frequently cited as the factor with the strongest association with the probability of churn [4]. Other important features include estimated salary, credit score, and account balance [9, 4].

Interestingly, customers who churn often exhibit higher estimated incomes and average balances [9]. This counterintuitive observation suggests that banks might be neglecting their high-value segments, leading to their attrition. This "churn paradigm" in high-value customers highlights that retention strategies cannot be one-size-fits-all.

A customer's product count is another significant predictive factor; customers with more products tend to be more loyal [8, 4]. In contrast, features such as credit card ownership may show a weaker correlation with churn [4].

Typical preprocessing steps include cleaning, encoding categorical variables in numerical values, and scaling numerical features [6, 7]. A common characteristic of churn datasets is class imbalance, where the number of non-churning customers significantly outweighs those who do (e.g., 20% churn in the Kaggle dataset [6, 8]). This imbalance in the data requires novel techniques, such as SMOTE (Synthetic Minority Oversampling Technique), which enables equalizing the class distribution and preventing models from biasing predictions towards the majority class [14, 7, 9]. Feature engineering, which involves creating new derived features such as BalanceSalaryRatio, TenureByAge, and CreditScoreGivenAge, is also crucial for improving model performance [6].

1.5. Synthesis and Emerging Trends

The literature review reveals clear trends in the field's evolution. Ensemble methods, particularly Random Forest and Gradient Boosting variants, have emerged as the dominant approaches, consistently outperforming traditional models. The ability of these types of algorithms to handle the non-linearity and complexity inherent in customer behavior data probably explains this superiority.

The growing importance of interpretability reflects a maturation in the field, where predictive accuracy must be balanced with model understanding. This is particularly relevant in the banking context, where automated decisions require justification and transparency for regulatory compliance.

The identification of counterintuitive patterns, such as higher churn among high-value customers, underscores the need for more sophisticated analyses that extend beyond traditional intuitions. These findings suggest opportunities for more nuanced and segmented retention strategies.

Finally, effective management of class imbalance has been established as a fundamental requirement for developing successful churn prediction models, with techniques such as SMOTE becoming standard practice in the field.

1.6. Contribution and Differentiation of the Study

Although our findings confirm the superiority of ensemble methods, the main contribution of this study is its methodological rigor and practice-oriented conclusions. Unlike other works, we employ advanced optimization tools, such as Optuna [15] and TPOT [16] for a more systematic and exhaustive hyperparameter search. Furthermore, we conduct a comprehensive comparison of strategies for handling class imbalance, evaluating not only

oversampling techniques like SMOTE/ADASYN but also native classifiers from `imbalanced-learn` [17] and `scikit-learn` [18].

The robustness of our findings is reinforced by a dual validation strategy, as demonstrated by the high correlation between our local AUC and the AUC from the Kaggle competition, which ensures the generalizability of our models. Ultimately, our primary contribution is not a single "best model" but a practical decision-making framework. This framework offers a choice: select the LGBM model to maximize predictive power via AUC, or the Balanced Random Forest to maximize minority class identification via Recall. This approach allows institutions to align model selection with their specific business objectives. Furthermore, the high interpretability of these models, explained using SHAP [19], provides additional support for the decision-making process.

2. Materials and Methods

The methodological structure proposed in this study is visually detailed in Figure 1.

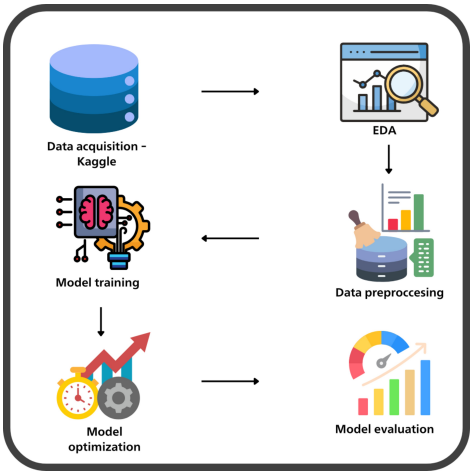


Figure 1: Proposed methodological structure for the construction of predictive models.

2.1. Dataset Description

The analysis was performed on the dataset from the Kaggle "Bank Customer Churn Prediction" challenge [20]. An initial exploratory data analysis

was performed to understand the structure, distributions, and relationships of the dataset.

The dataset consists of 15,000 observations and 14 variables. The variables were categorized as follows:

- Numerical: `CreditScore`, `Age`, `Tenure`, `Balance`, `EstimatedSalary`.
- Categorical: `NumOfProducts`, `Geography`, `Gender`, `HasCrCard`, `IsActive Member`, `Exited`.
- Unique Identifiers: `id`, `CustomerId`, `Surname`.

The unique identifier variables were removed as they provide no relevant information for the analysis. No missing values were found in any variable.

2.1.1. Descriptive statistics

A summary of the numerical variables is shown in Table 2, including measures of central tendency and dispersion for the 15,000 records.

The mean value of the CreditScore variable is 658.85, and its standard deviation is 72.59. The average age is 37.89 years, with a range of 18 to 74 years, indicating a predominantly adult clientele. The average Tenure of customers at the institution is approximately 5 years. A notable finding is in the Balance, where the median is 0.00, suggesting that a significant portion of customers have no balance in their accounts. However, the mean value is 43,124.06, driven by customers with considerably high balances. On average, customers have 1.59 products, and the mean EstSalary (Estimated Salary) is 118,134.81, also indicating significant variability.

Table 2: Descriptive statistics of numerical variables

Statistic	**CreditScore**	**Age**	**Tenure**	**Balance**	**NumProducts**	**EstSalary**
Observations	15,000	15,000	15,000	15,000	15,000	15,000
Mean	658.85	37.89	5.03	43 124.06	1.59	118 134.81
Std. Dev.	72.59	8.26	2.81	59 777.87	0.54	46 803.11
Min	431.00	18.00	0.00	0.00	1.00	11.58
25% (Q1)	602.00	32.00	3.00	0.00	1.00	83 629.60
Median	662.00	37.00	5.00	0.00	2.00	123 587.80
75% (Q3)	709.00	42.00	7.00	109 650.98	2.00	157 558.50
Max	850.00	74.00	10.00	187 530.66	4.00	1 557 802.00
Z-score	15 (0.10%)	150 (1%)	0 (0%)	0 (0%)	0 (0%)	1 (0.01%)

Table 3 summarizes the categorical data, highlighting key distributions within the customer base. The analysis shows a strong concentration of customers in France, with 8,971 individuals. The sample is predominantly male, with 8,358 records. Key financial behaviors include widespread credit card ownership, reported for 11,748 customers, and active membership status for 7,503 customers. In terms of product engagement, the most common scenario involves having two products. Critically, the data indicates a high retention rate, as the Exited variable shows that 11,942 out of the total customers have not left the bank.

Table 3: Descriptive Statistics of Categorical Variables

Variable	**Unique**	**Top**	**Freq**
Geography	3	France	8971
Gender	2	Male	8358
HasCrCard	2	True	11748
IsActiveMember	2	True	7503
NumOfProducts	4	2	8305
Exited	2	False	11942

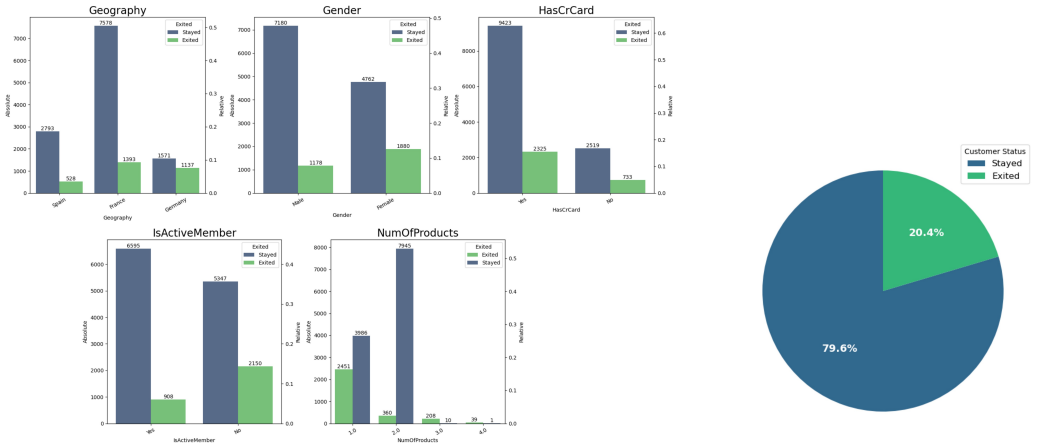
The distribution of these categorical features is visualized in Figure 2a.

The target or dependent variable, `Exited`, indicates whether a customer has churned. As shown in Figure 2b, the dataset exhibits a significant class imbalance with 11,942 non-churning customers and 3,058 churning customers. These figures correspond to 79.6% and 20.4% of the data, respectively, creating a ratio of approximately 4:1. This imbalance is a critical factor in the modeling phase.

2.1.2. Outlier analysis

An outlier analysis was conducted using the Z-score method with a threshold of $|Z| > 3$. The results in Table 2 indicate a low proportion of outliers, as all variables have a percentage better than 1%, suggesting that the data are mostly within a normal distribution and no aggressive cleaning will be required.

To complement the outlier analysis, the distributions and presence of outliers were also visualized using boxplots, as shown in Figure 3, which reveal that the churning population tends to have a higher median age and account balance.



(a) (b)

Figure 2: Distributions of the variables in the dataset. a) Distribution of categorical variables. b) Pie chart of the target categorical variable, `Exited`.

Specifically, the interquartile range (IQR) for Age in the churned group is shifted noticeably upward compared to the group that stayed. A similar, even more pronounced, separation is visible in the Balance variable, where the median balance for customers who exited is substantially higher. This suggests that these customers may be more financially established or are seeking better investment opportunities elsewhere.

In stark contrast, the distributions for CreditScore, EstimatedSalary, and Tenure appear remarkably similar between the two groups. For these variables, the medians and IQRs are nearly identical, indicating that a customer's credit score, estimated salary, or their length of service with the bank are not strong differentiating factors for churn when viewed in isolation.

The distributions of the numerical variables were examined through histograms (Figure 4). CreditScore follows a near-normal distribution, Age is right-skewed, Tenure is almost uniform, Balance is bimodal with a large number of zero balances, and EstimatedSalary is largely uniform.

2.1.3. Analysis of Linear Relationships

A linear correlation analysis was performed among the numerical variables of the dataset by calculating the Pearson correlation coefficient. The formula

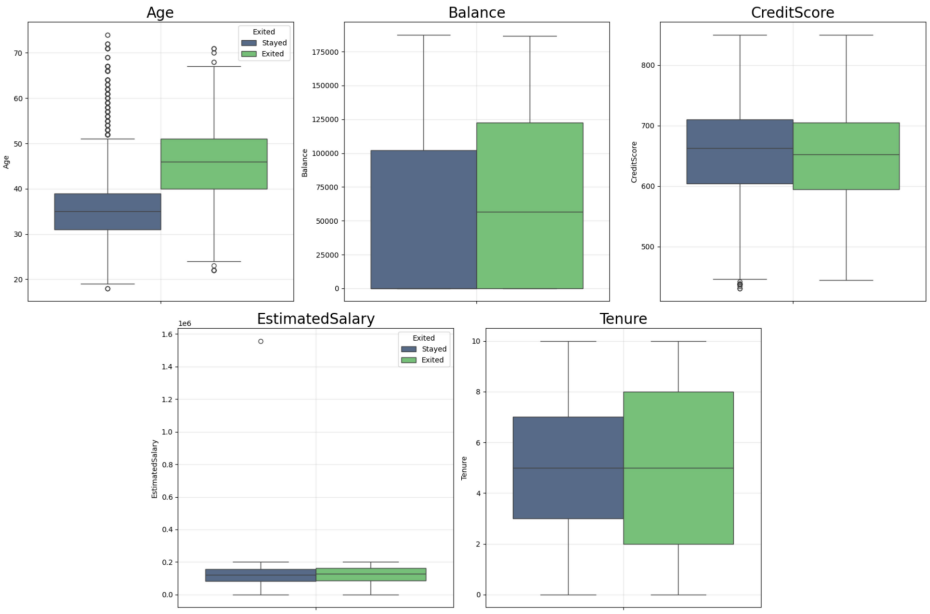


Figure 3: Boxplots of numerical variables grouped by churn status.

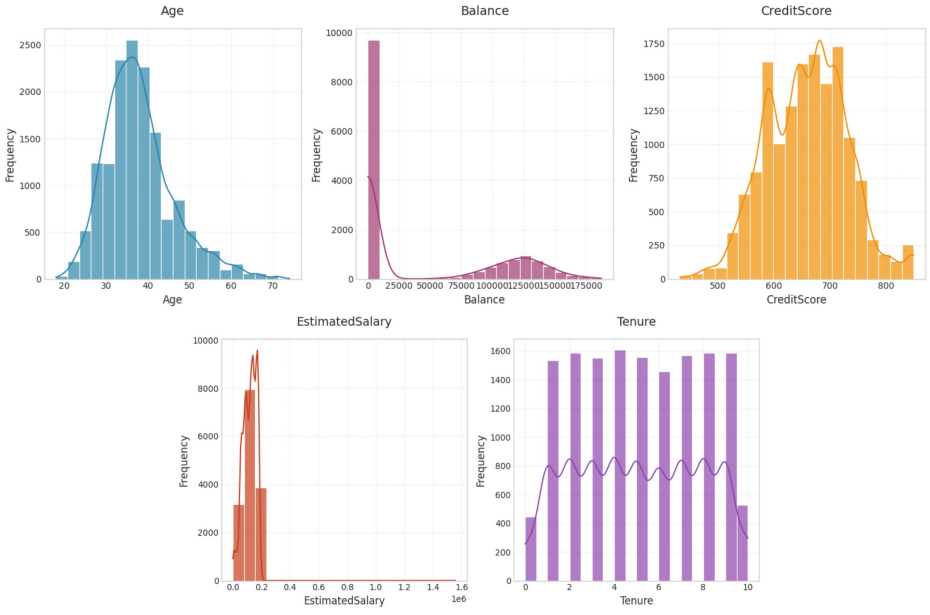


Figure 4: Histograms of numerical variables.

for this coefficient for a sample is given by:

$$r_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

where:

- cov(X,Y): Is the covariance between variables X and Y. Measures the direction of the linear relationship between the two variables.
- σ_X: Is the standard deviation of variable X. Measures the dispersion or variability of the data for X.
- σ_Y: Is the standard deviation of variable Y. Measures the dispersion or variability of the data for Y.

This analysis enables us to identify linear relationships between pairs of variables, which is useful for detecting potential redundancies or significant associations that could influence the prediction model.

To interpret the results, a threshold of 0.8 was established to identify strong correlations, a common methodological practice for assessing the magnitude of the relationship and preventing multicollinearity in predictive models [21, 22]. Figure 5 displays the correlation matrix represented as a heatmap. The matrix shows that no value exceeds this limit in absolute terms. The most significant relationship is observed between Exited and Age, with a correlation coefficient of 0.41. According to the interpretation scale [21], this value is considered moderate, indicating that as age increases, the likelihood of churn also increases. A weak negative correlation with a coefficient of 0.16 is also observed between Exited and Balance. Consequently, the variables are largely independent, reducing the risk of multicollinearity in predictive models.

2.2. Preprocessing

Before model training, several data preparation steps were executed:

- **Categorical Variable Encoding:** The Geography variable, having multiple categories, was transformed using *one-hot encoding*. The Gender variable was binary encoded, with male mapped to 0 and female to 1.

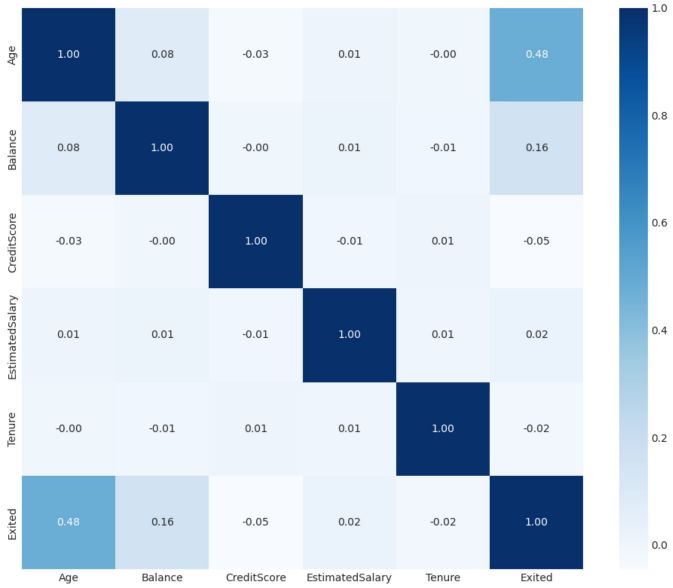


Figure 5: Correlation matrix of numerical variables.

- **Numerical Feature Scaling:** Numerical variables were standardized using `StandardScaler` for models sensitive to feature scale, such as Logistic Regression. For tree-based models (e.g., XGBoost and Random Forest), which are scale-invariant, the original unscaled data was used.
- **Handling Class Imbalance:** The significant class imbalance was addressed through a dual approach. First, oversampling techniques such as SMOTE and ADASYN were explored to balance the class distribution for standard classifiers. Second, specialized classifiers from the `imbalanced-learn` library, designed to handle imbalanced datasets natively, were evaluated.
- **Data Splitting:** The dataset was partitioned for training and testing using a stratified 80/20 split.

2.3. Modeling and Optimization

A diverse set of supervised classification algorithms was selected to address the churn prediction problem, based on their prevalence in the literature and varied approaches to classification. The models are grouped into two categories:

1. **Standard Classification Models:** This group includes a range of algorithms from a simple linear baseline to advanced tree-based ensembles known for their high predictive performance [3, 4, 7, 10, 8, 1, 9].
    - Extra Trees (Extremely Randomized Trees)
    - Gradient Boosting
    - Logistic Regression
    - Random Forest
    - XGBoost & LightGBM
2. **Specialized Models for Imbalanced Data:** To directly handle the class imbalance that commonly occurs in churn datasets [6, 8], classifiers from the `imbalanced-learn` library were selected as an alternative to applying techniques like SMOTE [7, 9].
    - Balanced Bagging Classifier (BBC)
    - Balanced Random Forest Classifier (BRF)
    - Easy Ensemble Classifier (EEC)
    - RUSBoost Classifier

The key characteristics of each model are summarized in Table 4.

2.4. Evaluation Metrics

To evaluate model performance, a set of metrics derived from the confusion matrix which tabulates True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) was employed. The key metrics used were:

- **Accuracy:** The proportion of correct predictions. It is obtained as $(TP + TN)/(TP + TN + FP + FN)$.
- **Precision:** Is the fraction of positive predictions that were actually correct. Its formula is $TP/(TP + FP)$.
- **Sensitivity (Recall):** The ability to correctly identify positive cases (churn). It is calculated as $TP/(TP + FN)$.
- **Specificity:** The ability to correctly identify negative cases (no-churn). Its formula is $TN/(TN + FP)$.

Table 4: Selected models and their characteristics

Model	Type	Requires Scaling	Imbalance Support
Logistic Regression	Linear	Yes	No
Random Forest	Trees	No	Partial
Gradient Boosting	Boosting	No	Partial
XGBoost	Boosting	No	Yes
LGBM	Boosting	No	Yes
Extra Trees	Randomized Trees	No	Partial
Easy Ensemble Classifier	Boosting + Undersampling	No	Yes
RUSBoost Classifier	Boosting + Undersampling	No	Yes
Balanced Bagging Classifier	Bagging + Undersampling	No	Yes
Balanced Random Forest	Randomized Trees	No	Yes

- **F1-Score:** The harmonic mean of precision and sensitivity, useful for imbalanced classes. It is calculated as $2\times(\text{Precision}\times\text{Sensitivity})/(\text{Precision}+\text{Sensitivity})$.
- **G-mean:** The geometric mean of sensitivity and specificity, robust to class imbalance. Its formula is $\sqrt{\text{Sensitivity} \times \text{Specificity}}$.

Additionally, the model's overall discriminatory power was evaluated using the AUC-ROC metric, which measures the model's ability to distinguish between classes. An AUC value close to 1 indicates superior performance.

3. Results

The following section describes the selected algorithms, outlines the validation approach used, and finally compares their performance based on various metrics.

3.1. Description of the approaches used

Several supervised classification models were selected, which were addressed using two approaches to identify the optimal configuration.

- **TPOT:** This AutoML framework, based on genetic algorithms, was used to automatically search for the best pipeline (model + preprocessing + hyperparameters). TPOT incorporates preprocessing strategies, feature selection, and class balancing as part of its evolutionary process.

- **Optuna:** Unlike an exhaustive search, this framework was used to implement an intelligent optimization process. Using a Bayesian sampler (TPE), Optuna proposes promising hyperparameter combinations based on previous results and, through the *pruning* technique, early discards low-performing trials to accelerate the search. This approach was applied to tune two families of models:
  - First, standard classification models (such as XGBoost, LightGBM, etc.) were optimized, exploring the application of oversampling techniques SMOTE and ADASYN as part of the process.
  - Second, the same optimization process was applied to tune the hyperparameters of specialized models from the `imbalanced-learn` library, which already includes native mechanisms for handling imbalanced data.

  Table 5 summarizes the optimized hyperparameters for both groups of models.

Additionally, two different strategies were carried out using Optuna to address the class imbalance. In the first strategy, SMOTE and ADASYN oversampling techniques were integrated directly into the optimization process, allowing the hyperparameter search to select the best model while considering the impact of balancing on performance. In the second strategy, the model was trained with the best hyperparameters obtained by Optuna on the original (unbalanced) data. Subsequently, the SMOTE and ADASYN techniques were applied independently to evaluate whether they contributed to an improvement in the performance of the final model.

In the case of TPOT, the generated models inherently incorporated class-balancing mechanisms as part of the automatically constructed pipeline.

The main characteristics of the baseline models evaluated were summarized in Table 4.

3.2. Model validation

To have one set of data for training and another for testing of the models, a stratified partition was performed, with 80% allocated for training and 20% for testing. This local split was maintained constant throughout the entire development and optimization phase to ensure consistency of the results. The search for the optimal model focused on two main strategies: hyperparameter optimization with Optuna and pipeline search with TPOT.

Model	Optimized hyperparameters
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, bootstrap
Gradient Boosting	n_estimators, learning_rate, max_depth, min_samples_split, min_samples_leaf, subsample
XGBoost	n_estimators, learning_rate, max_depth, min_child_weight, subsample, colsample_bytree, gamma
LGBM	n_estimators, learning_rate, max_depth, min_child_samples, subsample, colsample_bytree, reg_alpha, reg_lambda
Logistic Regression	C, penalty, solver, class_weight
Extra Trees	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
Easy Ensemble	n_estimators
RUSBoost	n_estimators, learning_rate, max_depth
Balanced Bagging Classifier	n_estimators, max_samples, max_features, max_depth
Balanced Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features

Table 5: Optimized hyperparameters per model during the search with Optuna.

During optimization with Optuna, the local training set was used to tune the hyperparameters of each model. To ensure that the 250-trial optimization process remained computationally feasible, a single train-validation split of this data was used to evaluate the performance of each hyperparameter set. The metric AUC-ROC in the validation set was used as the primary metric to be optimized. The search space was broad, covering not only the optimization of the base models listed in Table 5, but also the impact of oversampling techniques (SMOTE and ADASYN). Furthermore, the same optimization process was applied to models from the `imbalanced-learn` library, which are specifically designed to handle imbalanced data natively.

Finally, to obtain a comprehensive performance assessment, a dual AUC evaluation approach was used. For the local test set, metrics such as accuracy, recall, F1-score, G-mean, and the Local AUC (AUC_L) were calculated. For the external evaluation, the Kaggle AUC (AUC_K) was determined through the competition platform. The optimized model then generated predictions on the official dataset, which were then submitted to the platform to obtain the final score.

In the case of TPOT, the framework applied its evolutionary process

on the training set to find the optimal model. This automated search was configured for 60 `generations` with a `population_size` of 20, running for a maximum of 30 minutes (`max_time_mins`). Internal validation was performed using 5-fold cross-validation (`cv`), and for reproducibility, a `random_state` of 52 was set. Subsequently, the best model identified by this process was evaluated under the same dual metric scheme to ensure a fair comparison.

3.3. Model results

Table 6 details the hyperparameters of the best models obtained using TPOT, Optuna (without and with integrated balancing), and Imblearn, respectively. This presentation enables the visualization of the differences in the final configuration of each model, resulting from the automated search approaches employed in both cases.

On the other hand, Table 7 presents a comparison of the performance achieved by the selected models in the test set. To visualize their discrimination capability, the model performance is illustrated through ROC curves, which are presented in two groups based on performance. Figure 7a shows the models with the highest performance, where top models such as LGBM (TPOT) achieve an AUC of 0.937. In contrast, Figure 7b groups the models with lower performance, with results ranging from an AUC of 0.932 for RandomForest down to 0.877 for LogisticRegression. For a more granular breakdown, Figure 6 displays the confusion matrix for each of the five best-performing models. From these matrices, a key insight emerges: the Gradient Boosting model (Figure 6d) was the most effective at identifying customers who would actually churn, correctly flagging 434 cases. Together, these metrics and visualizations enable an analysis that extends beyond overall accuracy to assess the ability to distinguish between classes in an imbalanced scenario.

4. Discussion

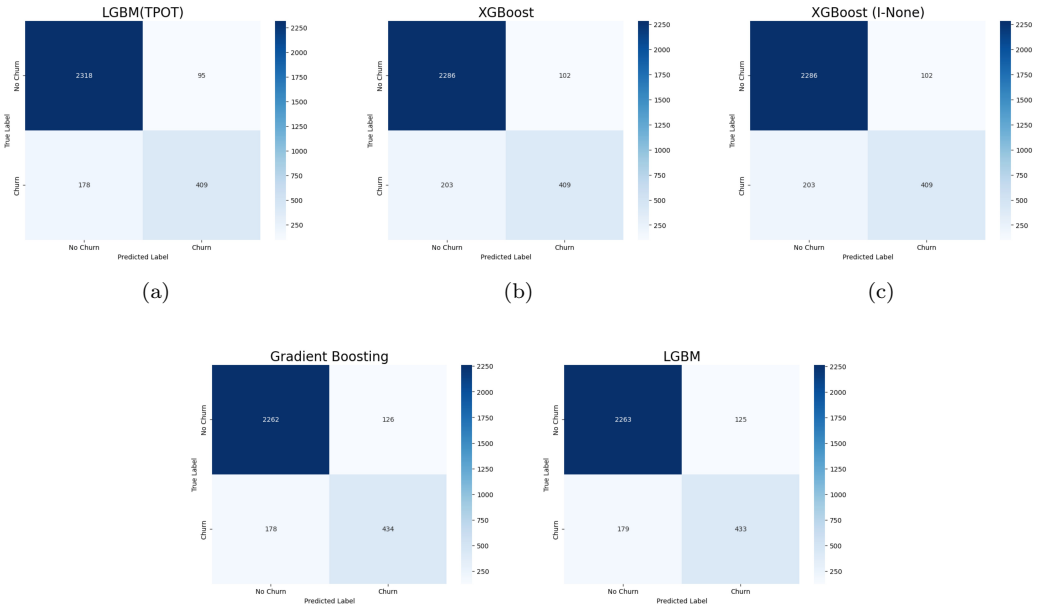
The analysis of our results reveals nuanced findings of significant practical value. A primary finding is the high overall predictive power of the evaluated models, with most classifiers exceeding an AUC of 0.93 in both local validation and the Kaggle competition. This consistency between the local and Kaggle AUC scores provides a robust validation of our experimental methodology, confirming that the models do not suffer from overfitting and possess excellent generalization capabilities.

Model	Hyperparameter	Range	Best values
LGBM (TPOT)	max_depth	–	9
	n_estimators	–	55
	num_leaves	–	8
XGB	n_estimators	100-400	200
	learning_rate	0.01-0.3	0.1107
	max_depth	3-10	9
	min_child_weight	1-10	10
	subsample	0.7-1.0	0.8266
	colsample_bytree	0.7-1.0	0.7236
	gamma	0-5	4.5140
LGBM (ADASYN)	sampler	None, SMOTE, ADASYN	ADASYN
	n_estimators	100-400	157
	learning_rate	0.01-0.3	0.0532
	max_depth	3-10	6
	min_child_samples	1-20	17
	subsample	0.7-1.0	0.9669
	colsample_bytree	0.7-1.0	0.7485
	reg_alpha	0-2	0.6489
	reg_lambda	0-2	0.8733
BRF	n_estimators	50-300	181
	max_depth	3-20	7
	min_samples_split	2-20	6
	min_samples_leaf	1-10	9
	max_features	sqrt, log2, None	None

Table 6: Range and best values of the hyperparameters for the best models obtained with TPOT, Optuna, and Imblearn techniques. For the LGBM (TPOT) model, no search ranges are specified, as its hyperparameters are determined through an automated optimization process.

Model	Type	Acc.	AUC K	AUC L	Prec.	Recall	F1	G-Mean	Avg.
TPOT (LGBM)	I	0.910	0.940	**0.937**	0.820	0.690	0.750	0.816	0.828
Optuna (XGB)	N	0.909	0.940	0.935	0.810	0.700	0.750	0.818	0.828
Optuna (XGB) + SMOTE	P	0.912	0.938	0.935	0.800	0.730	0.760	0.837	0.845
Optuna (XGB) + ADASYN	P	0.912	0.938	0.935	0.800	0.740	**0.770**	0.839	0.846
Optuna (RF)	I	0.893	0.936	0.932	0.810	0.621	0.700	0.773	0.792
Optuna (GB)	I	0.899	0.939	0.936	0.780	0.709	0.740	0.820	0.828
Optuna (XGB)	I	0.898	0.941	0.936	0.800	0.668	0.730	0.800	0.813
Optuna (LGBM)	I	0.899	**0.942**	0.935	0.780	0.708	0.740	0.819	0.828
Optuna (LR)	I	0.818	0.891	0.877	0.540	0.796	0.640	0.810	0.810
Optuna (ExtraTrees)	I	0.889	0.935	0.931	**0.820**	0.588	0.680	0.754	0.777
Imblearn (EEC)	I	0.854	0.929	0.924	0.600	0.815	0.690	0.839	0.840
Imblearn (RUS)	I	0.801	0.900	0.889	0.510	0.778	0.610	0.792	0.792
Imblearn (BBC)	I	0.876	0.938	0.930	0.660	0.828	0.730	0.858	0.858
Imblearn (BRF)	I	0.863	0.939	0.932	0.620	**0.856**	0.720	**0.861**	**0.861**

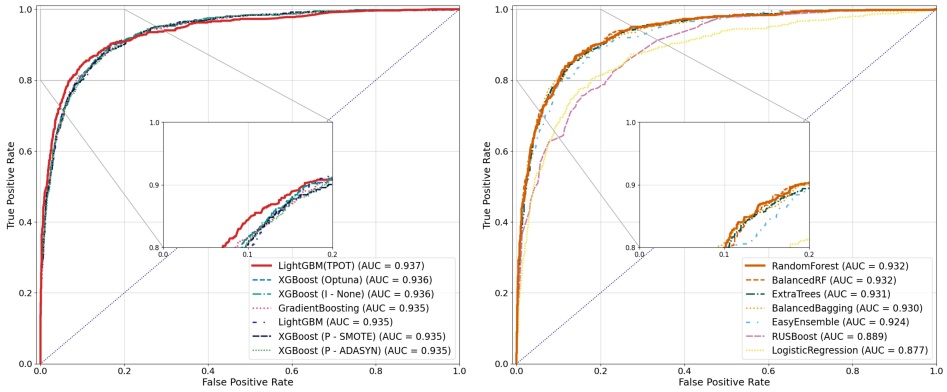
Table 7: Comparison of performance metrics. The column 'Type' indicates how the balancing technique was applied: N = No technique, P = Applied post-optimization, I = Integrated into optimization.



(a) (b) (c)

(d) (e)

Figure 6: Confusion matrices for the five best performing models



(a) Models with higher performance (b) Models with lower performance

Figure 7: Comparative ROC curves of the evaluated models, grouped by performance

A comparison with the existing literature, summarized in Table 1, shows that our best models achieve state-of-the-art performance. For example, our LGBM model optimized with Optuna reached a Kaggle AUC of 0.941, outperforming many previously reported implementations. This result highlights

the effectiveness of automatic hyperparameter optimization approaches, such as Optuna, which efficiently explore complex search spaces compared to manual or grid searches.

However, the most significant contribution of this study is not the crowning of a single model, but the demonstration of a clear *trade-off* between global predictive accuracy and the ability to correctly identify the minority class. The LGBM model, like other *boosting* algorithms, excels in AUC optimization because its goal is to finely calibrate probabilities across the entire decision spectrum. In contrast, the BRF classifier emerges as the superior choice in metrics such as Recall (0.856) and G-Mean (0.861). Its design, which inherently balances the data during the construction of each tree, makes it exceptionally effective for the business objective of identifying as many customers at risk of churn as possible, even if this means sacrificing a small fraction of overall accuracy.

4.1. Model Interpretability

To better understand the decisions of the model optimized with Optuna, the SHAP interpretability method was applied. This approach decomposes each customer's prediction into individual contributions from its features. Figure 8 presents the summary plot, which provides a global overview of the impact of each feature on the model predictions.

As can be observed, features such as NumOfProducts, Age, and IsActiveMember play a decisive role in churn behavior. The color intensity represents the feature value (high or low), while the position on the horizontal axis indicates whether the feature drives the prediction towards churn (positive) or retention (negative).

In addition to the global analysis, a concrete customer example was examined to illustrate how the model makes decisions. Figure 9 breaks down the factors influencing the prediction for a specific client. The analysis reveals that the model's decision is overwhelmingly driven by a single high-risk factor: the customer holding only one product (NumOfProducts = 1), which contributes a SHAP value of +1.35 to the probability of churn.

Although there are important features suggesting loyalty, such as the customer's age (Age = 34) and active membership status (IsActiveMember = 1), which exert significant opposing forces (SHAP values of -0.97 and -0.49, respectively), these are not sufficient to overturn the prediction. The substantial negative impact of having a single product outweighs these positive influences. Secondary factors, such as a zero balance (Balance = 0), slightly

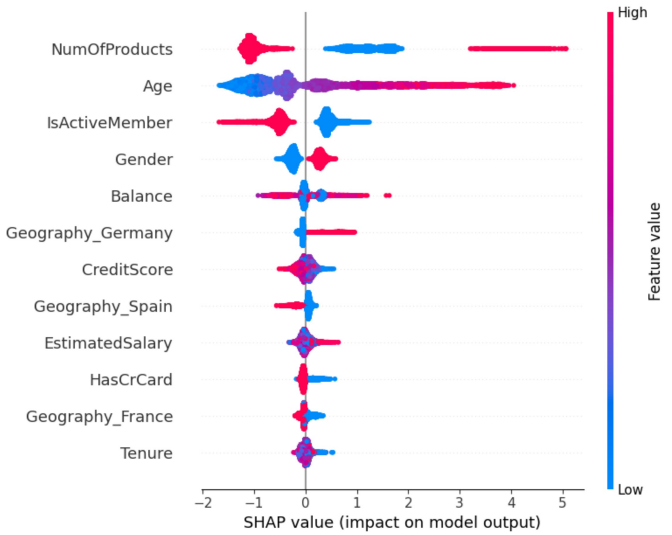


Figure 8: SHAP summary of the LGBM model optimized with Optuna, showing the highest AUC K.

reinforce the customer churn prediction, leading the model to conclude that despite the presence of several retention indicators, the risk of exit remains high.

4.2. Practical Implications for the Banking Sector

The results of this study provide a clear guide for financial institutions to implement data-driven retention strategies, thereby optimizing their resources.

- **Targeted retention strategies:** The interpretability analysis identifies the number of products, the age, and the active membership status as the main predictors of churn. This enables banks to segment high-risk customers and design concrete actions, such as cross-selling campaigns for clients with only one product, personalized services for older customers, or reactivation initiatives for inactive members.

- **Model alignment with business objectives:** The study shows that the choice of model should be aligned with the strategic goal. To achieve maximum overall predictive accuracy (AUC), the optimized LGBM model is superior. However, if the objective is to maximize the

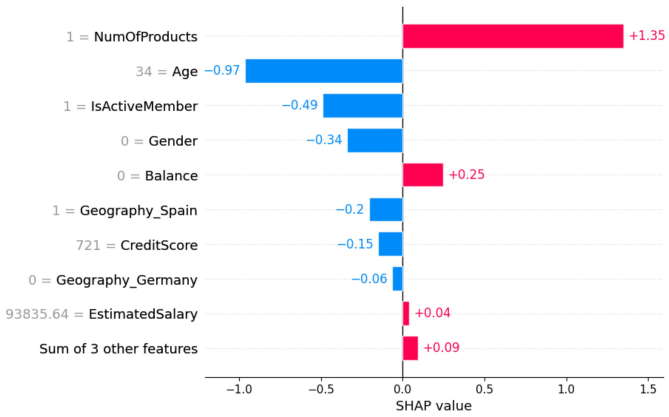


Figure 9: Individual example of SHAP explanation.

identification of at-risk customers so that retention teams can intervene, the BRF classifier is the preferred option due to its higher Recall.

- **Resource optimization:** By assigning a *churn* probability to each customer, banks can focus their efforts and budget on the segments with the highest risk of leaving. This proactive approach is more efficient and cost-effective than broad retention campaigns.

5. Conclusions

This study addressed the challenge of predicting customer churn in the banking sector through a systematic and rigorous evaluation of multiple machine learning models. Through a process that included exhaustive preprocessing, advanced handling of class imbalance, and hyperparameter optimization using tools like Optuna and TPOT, the following key conclusions were drawn.

- There is no single "best model" but rather an optimal solution aligned with business objectives. The primary contribution of this work is a decision-making framework that enables financial institutions to select the most suitable model in accordance with their strategic priorities.

- For maximum predictive accuracy, Gradient Boosting models are the preferred choice. The LGBM model, finely tuned with Optuna, achieved the highest global discriminatory power, reaching an AUC of 0.941 on

the Kaggle platform, making it an ideal tool for correctly classifying the largest possible number of customers.

- For the priority identification of at-risk customers, specialized models for imbalanced data are superior. The BRF classifier demonstrated an outstanding ability to correctly identify customers who leave the bank, achieving the highest Recall. This model is the recommended option for retention teams that need to maximize opportunities for intervention..

- Model interpretability provides strategic and actionable information. The analysis using SHAP identified `NumOfProducts`, `Age`, and `IsActive Member` as the most significant factors in determining churn. These findings enable banks to transition from predictive analysis to the implementation of proactive and segmented retention strategies, such as loyalty campaigns for older customers or cross-selling initiatives for those with a single product.

Statements and Declarations

5.1. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) only used AI to improve the readability and clarity of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and assume(s) full responsibility for the content of the published article.

5.2. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

5.3. Funding Information

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5.4. Author contributions

Anderson Joseph Ochoa Trujillo: Validation, Writing – review & editing,

Cristian Andrés Villarreal Orozco: Conceptualization, Visualization, Methodology, Software & Writing,

Juan Pablo Hoyos Sanchez: Conceptualization, Writing – review & editing.

5.5. Data Availability Statement

The source code developed for this research is available in the GitHub repository: Bank-Churn-ML

5.6. Research Involving Human and /or Animals

Not Applicable

5.7. Informed Consent

Not Applicable

References

[1] L. W. Widianti, A. S. B. Karno, W. Hastomo, A. N. Utomo, D. Arif, I. S. K. Wardhana, D. Strydom, Improved banking customer retention prediction based on advanced machine learning models, Indonesian Journal of Information Systems 7 (2) (2025) 178–193. doi:10.24002/ijis.v7i2.10364.
URL https://ojs.uajy.ac.id/index.php/IJIS/article/view/10364

[2] E. Kabbar, N. Herath, Customer churn prediction to enhance customer retention strategies in the banking industry: A study using seven machine learning algorithms, Journal of Software and Systems Development 2025 (2025) 1–10, article ID 786386. doi:10.5171/2025.786386.
URL https://doi.org/10.5171/2025.786386

[3] P. A. Jufin, N. Amrutha, Bank customer churn prediction, Indian Journal of Data Mining 2 (2) (2023) 1–5.
URL https://www.researchgate.net/publication/376983037_Bank_Customer_Churn_Prediction

[4] H. Zhu, Bank customer churn prediction with machine learning methods, Advances in Economics, Management and Political Sciences 69, published on 8 January 2024 (2024). doi:10.54254/2754-1169/69/20230773.
URL https://doi.org/10.54254/2754-1169/69/20230773

[5] A. Khaliq, et al., From data to decisions: Predictive machine learning models for customer retention in banking, ResearchGate (2024).
URL https://www.researchgate.net/publication/384995568_From_Data_to_Decisions_Predictive_Machine_Learning_Models_for_Customer_Retention_in_Banking

[6] Z. Haidar, Binary-classification-with-a-bank-churn-dataset, GitHub repository (2024).
URL https://github.com/zainhaidar16/Binary-Classification-with-a-Bank-Churn-Dataset

[7] B. A. Maulana, N. Hidayati, Churn prediction in credit customers using random forest and xgboost methods, Indonesian Journal of Data and Science 6 (1) (2025) 81–89. doi:10.56705/ijodas.v6i1.215.
URL https://www.jurnal.yoctobrain.org/index.php/ijodas/article/view/215

[8] R. Ashraf, Bank customer churn prediction using machine learning framework, J Appl Financ Bank 14 (4) (2024) 65–109.
URL https://www.scienpress.com/Upload/JAFB/Vol%2014_4_5.pdf

[9] T. Zhang, Prediction and clustering of bank customer churn based on xgboost and k-means, BCP Business & Management 23 (2022) 360–366. doi:10.54691/bcpbm.v23i.1373.
URL https://bcpublication.org/index.php/BM/article/view/1373

[10] E. Kabbar, N. Herath, Customer churn prediction to enhance customer retention strategies in the banking industry: A study using seven machine learning algorithms, Journal of Software & Systems Development 2025 (2025). doi:10.5171/2025.786386.
URL https://doi.org/10.5171/2025.786386

[11] X. Liu, et al., Machine learning based customer churn prediction in banking sector, ResearchGate (2024).
URL https://www.researchgate.net/publication/384203868_Machine_Learning_Based_Customer_Churn_Prediction_in_Banking_Sector

[12] P. P. Singh, F. I. Anik, R. Senapati, A. Sinha, N. Sakib, E. Hossain, Investigating customer churn in banking: a machine learning approach and visualization app for data science and management, Data Science and Management 7 (1) (2024) 7–16. doi:https://doi.org/10.1016/j.dsm.2023.09.002.
URL https://www.sciencedirect.com/science/article/pii/S2666764923000401

[13] Kaggle, Binary classification with a bank churn dataset, Kaggle Competition (2024).
URL https://www.kaggle.com/competitions/playground-series-s4e1/data

[14] J. Dunstan, F. Villena, J. Hoyos, V. Riquelme, M. Royer, H. Ramírez, J. Peypouquet, Predicting no-show appointments in a pediatric hospital in Chile using machine learning, Health Care Management Science 26 (2) (2023) 313–329. doi:10.1007/s10729-022-09626-z.
URL https://doi.org/10.1007/s10729-022-09626-z

[15] T. Akiba, S. Sano, T. Yanase, T. Ohta, M. Koyama, Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining, 2019, pp. 2623–2631.

[16] R. S. Olson, J. H. Moore, TPOT: A Tree-Based Pipeline Optimization Tool for Automating Machine Learning, Springer International Publishing, Cham, 2019, pp. 151–160. doi:10.1007/978-3-030-05318-5_8.
URL https://doi.org/10.1007/978-3-030-05318-5_8

[17] G. Lemaître, F. Nogueira, C. K. Aridas, Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning, Journal of Machine Learning Research 18 (17) (2017) 1–5.
URL http://jmlr.org/papers/v18/16-365.html

[18] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, Édouard Duchesnay, Scikit-learn: Machine learning in python, Journal of Machine Learning Research 12 (85) (2011) 2825–2830.
URL http://jmlr.org/papers/v12/pedregosa11a.html

[19] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, Curran Associates, Inc., 2017.
URL http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model
pdf

[20] F. J. Adekunle, Bank customer churn prediction challenge, https://kaggle.com/competitions/bank-customer-churn-prediction-challenge, kaggle (2025).

[21] M. M. Mukaka, A guide to appropriate use of correlation coefficient in medical research, Malawi Medical Journal 24 (3) (2012) 69–71.
URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3576830/

[22] A. K. Ahmad, A. Jafar, K. Aljoumaa, Customer churn prediction in telecom using machine learning in big data platform, Journal of Big Data 6 (1) (2019). doi:10.1186/s40537-019-0191-6.
URL https://doi.org/10.1186/s40537-019-0191-6